Hand-out (Unit 4)

Example 1. Write a program for the 8051 to transfer letter 'A' serially at 4800- baud rate, 8 bit data, 1 stop bit continuously.

```
ORG 0000H
LJMP START
ORG 0030H
START: MOV TMOD, #20H
MOV TH1, #0FAH
MOV SCON, #50H
SETB TR1
AGAIN: MOV SBUF, #'A'
BACK: JNB TI, BACK
CLR TI
SJMP AGAIN
END
```

Example 2. Write a program for the 8051 to transfer the message 'EARTH' serially at 9600 baud, 8 bit data, 1 stop bit continuously.

```
ORG 0000H
LJMP START
ORG 0030H
START: MOV TMOD, #20H
MOV TH1, #0FDH        ; load count to get reqd. baud rate of 9600
MOV SCON, #50H        ; initialise uart in mode 2
SETB TR1
LOOP: MOV A, #'E'
ACALL LOAD
MOV A, #'A'
ACALL LOAD
MOV A, #'R'
ACALL LOAD
MOV A, #'T'
ACALL LOAD
MOV A, #'H'
ACALL LOAD
SJMP LOOP
LOAD: MOV SBUF, A
HERE: JNB TI, HERE
CLR TI
RET
END
```

<u>Homework</u>

1. What is auto reload mode of Timers in 8051?
2. Explain the pins of RS 232 DB9 connector.
3. Write an ALP for 8051 microcontroller to receive the data serially at the baud rate of 9600 and put them in the port P1. Do it in a loop.
4. Write the connection from 8051 microcontroller to RS 232 connector.
5. What are synchronous and asynchronous serial communication standards?